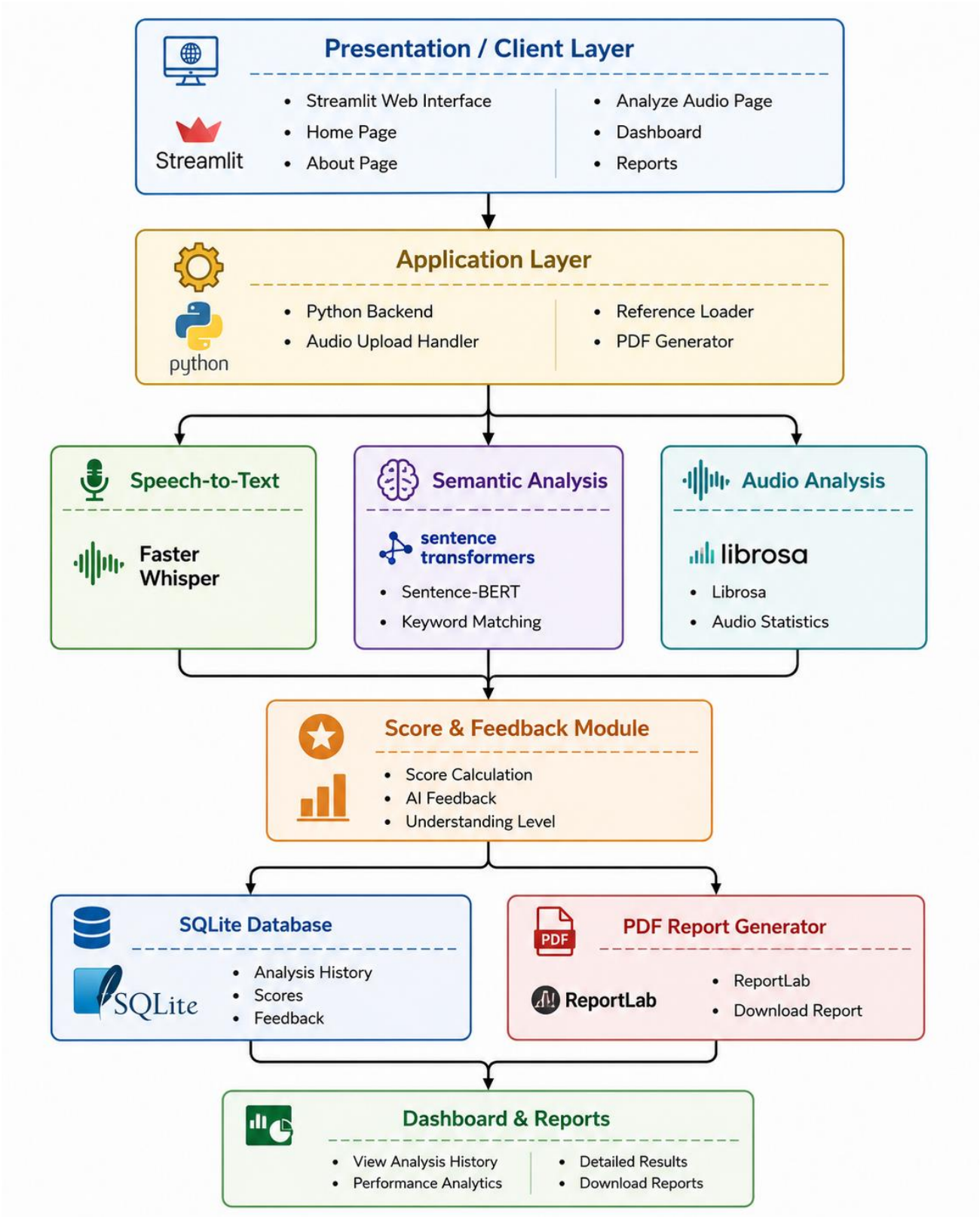
Solution Architecture

Date	12 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.


Presentation / Client Layer
Streamlit
Streamlit Web Interface
Home Page
About Page
Analyze Audio Page
Dashboard
Reports
Application Layer
python
Python Backend
Audio Upload Handler
Reference Loader
PDF Generator
Speech-to-Text
Faster Whisper
Semantic Analysis
sentence transformers
Sentence-BERT
Keyword Matching
Audio Analysis
librosa
Librosa
Audio Statistics
Score & Feedback Module
Score Calculation
AI Feedback
Understanding Level
SQLite Database
SQLite
Analysis History
Scores
Feedback
PDF Report Generator
PDF
ReportLab
ReportLab
Download Report
Dashboard & Reports
View Analysis History
Performance Analytics
Detailed Results
Download Reports

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the graphical user interface for uploading audio, viewing analysis, dashboards, and reports.	Streamlit, HTML, CSS
Application Layer	Manages application flow, user requests, file handling, and report generation.	Python
Speech-to-Text Module	Converts uploaded speech into text.	Faster Whisper
Semantic Analysis Module	Compares transcript with reference concepts and calculates semantic similarity.	Sentence-BERT
Keyword Matching Module	Detects matched and missing keywords from the explanation.	Python
Audio Analysis Module	Extracts waveform, duration, speech rate, energy, and silence percentage.	Librosa, SoundFile
Score & Feedback Module	Calculates overall score and generates AI-based feedback.	Python
Database Layer	Stores transcripts, scores, feedback, and analysis history.	SQLite
Report Generation Module	Generates downloadable PDF reports containing analysis results.	ReportLab